

Smart Home Environment & Security Monitoring System

```
#include <DHT.h>
```

```
#define DHTPIN 4
```

```
#define DHTTYPE DHT22
```

```
#define PIR_PIN 27
```

```
#define GREEN_LED 18
```

```
#define RED_LED 19
```

```
#define BUZZER 23
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    pinMode(PIR_PIN, INPUT);
```

```
    pinMode(GREEN_LED, OUTPUT);
```

```
pinMode(RED_LED, OUTPUT);
pinMode(BUZZER, OUTPUT);

dht.begin();

Serial.println("SMART HOME MONITOR");
}

void loop() {

float temp = dht.readTemperature();

bool motion = digitalRead(PIR_PIN);

digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, LOW);
digitalWrite(BUZZER, LOW);

String status = "SAFE";

if(temp > 35 && motion) {

status = "CRITICAL ALERT";

digitalWrite(RED_LED, HIGH);
digitalWrite(BUZZER, HIGH);
```

```
}
```

```
else if(temp > 35) {
```

```
    status = "HIGH TEMPERATURE";
```

```
    digitalWrite(RED_LED, HIGH);
```

```
}
```

```
else if(motion) {
```

```
    status = "INTRUDER DETECTED";
```

```
    digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(BUZZER, HIGH);
```

```
}
```

```
else {
```

```
    digitalWrite(GREEN_LED, HIGH);
```

```
}
```

```
Serial.print("Temperature: ");
```

```
Serial.print(temp);
```

```
Serial.println(" C");
```

```
Serial.print("Motion: ");  
Serial.println(motion ? "YES" : "NO");
```

```
Serial.print("Status: ");  
Serial.println(status);
```

```
Serial.println("-----");
```

```
delay(2000);
```

```
}
```

Wokwi Link

<https://wokwi.com/projects/467712447698029569>